FINAL PROJECT • PART I

Appendices

and supplementary material

The solutions to Questions 1–5 are complete. The following appendices develop supporting interpretations, circuit walkthroughs, and implementation details. Each discussion links to its numbered main-text application; the same references identify that application on paper.

Appendix A: Deferred measurement

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APPENDIX A

Question 1: Deferred measurement

A.1 Clause (a): Measurement without looking

A.1.i Why unread measurement removes coherence

This discussion expands the comparison of the input and output in Question 1(a) (Section 1.a.i). I use pure-state blocks to show why averaging the conditional density matrices removes coherence between the measurement alternatives while preserving the state within each alternative.

The two branches of a pure state. A general normalized pure state of the n original qubits can be grouped by the value of its first qubit as

$$|\psi\rangle = \alpha\,|0\rangle\,|\phi_0\rangle + \beta\,|1\rangle\,|\phi_1\rangle\,, \qquad |\alpha|^2 + |\beta|^2 = 1, \tag{A.1.1}$$

where $|\phi_0\rangle$ and $|\phi_1\rangle$ are normalized states of the remaining $n-1$ qubits for the branches whose coefficients are nonzero. A branch with zero coefficient is absent and its corresponding normalized vector may be chosen arbitrarily.

Forming the outer product $\rho = |\psi\rangle\langle\psi|$ produces four terms. Ordering the basis by the value of the first qubit displays those terms as the following 2×2 *block matrix*; each entry is itself an operator on the remaining $n-1$ qubits:

$$\rho = \begin{pmatrix} |\alpha|^2\,|\phi_0\rangle\langle\phi_0| & \alpha\overline{\beta}\,|\phi_0\rangle\langle\phi_1| \\ \overline{\alpha}\beta\,|\phi_1\rangle\langle\phi_0| & |\beta|^2\,|\phi_1\rangle\langle\phi_1| \end{pmatrix}. \tag{A.1.2}$$

The diagonal blocks lie entirely inside one measurement alternative. The two off-diagonal blocks are the coherence terms joining the 0 and 1 alternatives. They are adjoints of one another, so they are the two matrix orientations of one coherent relationship, not two independent physical coherences.

If the result is read, answer 0 occurs with probability $|\alpha|^2$ and leaves $|0\rangle\,|\phi_0\rangle$; answer 1 occurs with probability $|\beta|^2$ and leaves $|1\rangle\,|\phi_1\rangle$. When the result is not retained, we do not condition the output on either answer. The unread ensemble therefore averages the two conditional density matrices with their outcome probabilities:

$$\begin{aligned} \rho_{\text{unread}} &= |\alpha|^2\,|0\rangle\,|\phi_0\rangle\,\langle 0|\,\langle\phi_0| + |\beta|^2\,|1\rangle\,|\phi_1\rangle\,\langle 1|\,\langle\phi_1| \\ &= \begin{pmatrix} |\alpha|^2\,|\phi_0\rangle\langle\phi_0| & 0 \\ 0 & |\beta|^2\,|\phi_1\rangle\langle\phi_1| \end{pmatrix}. \end{aligned} \tag{A.1.3}$$

This average does not add conditional *amplitudes*, so it creates no cross terms. Comparing the two block matrices shows the change: the diagonal branch terms stay, while the off-diagonal coherence terms disappear.

A one-qubit example. For $n = 1$, take $|+\rangle = \big(|0\rangle + |1\rangle\big)/\sqrt{2}$. The input and unread output are

$$\rho_+ = |+\rangle\langle+| = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \mathcal{D}(\rho_+) = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \frac{I_2}{2}. \tag{A.1.4}$$

Both states give 0 and 1 with probability $1/2$ in the computational basis, so that measurement alone cannot reveal the difference. A measurement in the $\{|+\rangle\,, |-\rangle\}$ basis does reveal it:

$$\Pr(+\mid \rho_+) = \operatorname{Tr}\big(|+\rangle\langle+|\,\rho_+\big) = 1, \qquad \Pr\big(+\mid \mathcal{D}(\rho_+)\big) = \operatorname{Tr}\big(|+\rangle\langle+|\,I_2/2\big) = \frac{1}{2}. \tag{A.1.5}$$

The off-diagonal entries therefore have observable meaning: they allow the original alternatives to interfere when tested in another basis. Keeping only the classical frequencies of 0 and 1 does not preserve that information. For several qubits, the same statement applies to the two blocks selected by the first qubit. Coherence among states *inside* either block remains.

Return to the three circuits. In circuits (ii) and (iii), coherence can remain in the joint system and record. Tracing out the orthogonal record removes the cross terms from the original system alone.

Main-text reference: Section 1.a.i.

A.1.ii Phase kickback: an eigenphase becomes a relative phase

This discussion places the phase-basis record in circuit (iii) (Section 1.a.iii) within the general phase-kickback mechanism. The main proof follows the actual Hadamard and controlled-Z gates branch by branch and performs the circuit-specific partial trace. Here I abstract that calculation to a controlled unitary with an arbitrary eigenphase, then show how a final Hadamard can turn the phase into observable probabilities.

The general mechanism. Let U be a unitary on a target register T, and let $|u\rangle_T$ be a normalized eigenstate with $U|u\rangle_T = e^{2\pi i\theta}|u\rangle_T$, where $\theta \in [0,1)$. The ancilla A is the control. Keeping the target first and ancilla last in the tensor order, define the controlled operation by

$$C_A(U) := I_T \otimes (|0\rangle\langle 0|)_A + U \otimes (|1\rangle\langle 1|)_A. \tag{A.1.6}$$

For any normalized control state $c_0|0\rangle_A + c_1|1\rangle_A$, its action is

$$C_A(U)\left[|u\rangle_T (c_0|0\rangle_A + c_1|1\rangle_A)\right] = c_0|u\rangle_T|0\rangle_A + c_1 U|u\rangle_T|1\rangle_A = |u\rangle_T\left(c_0|0\rangle_A + c_1 e^{2\pi i\theta}|1\rangle_A\right). \tag{A.1.7}$$

The target factors out in the same eigenstate. Its eigenvalue has become the relative phase of the control's $|1\rangle_A$ amplitude. This is why the effect is called **phase kickback**. Applying U unconditionally to this fixed eigenstate would supply only a global phase; applying it on one component of a coherent control creates a phase difference between the control components.

In circuit (iii), $U = Z$. The target eigenstates $|0\rangle$ and $|1\rangle$ have eigenphases $\theta = 0$ and $\theta = 1/2$, respectively. With a $|+\rangle_A$ control, (A.1.7) therefore gives exactly the records $|+\rangle_A$ and $|-\rangle_A$. For a superposition of target eigenstates, the calculation is applied separately to each eigenstate and the resulting branches are added linearly, as in (1.a.25). Different eigenphases can then correlate the target with different control states; one cannot in general factor out a single unchanged target state.

How the phase can be read. A final Hadamard converts the relative phase into computational-basis probabilities. For the fixed eigenstate above, the control evolves as

$$\frac{|0\rangle_A + e^{2\pi i\theta}|1\rangle_A}{\sqrt{2}} \xrightarrow{H_A} \frac{1+e^{2\pi i\theta}}{2}|0\rangle_A + \frac{1-e^{2\pi i\theta}}{2}|1\rangle_A, \qquad \Pr(0) = \cos^2(\pi\theta). \tag{A.1.8}$$

For the two eigenphases of Z, the outcomes are deterministic: $|+\rangle$ becomes $|0\rangle$ and $|-\rangle$ becomes $|1\rangle$. Circuit (iii) discards the record, so it does not need this last conversion. Its orthogonal phase-basis records already suffice for the circuit-specific partial-trace calculation in the main proof. Phase estimation uses the same mechanism to learn an unknown eigenphase; the phase-estimation primer in Appendix C.1 follows that construction in detail.

Main-text reference: Section 1.a.iii.

A.1.iii General partial-trace identities

The main proof develops the ancilla coordinates and applies the partial trace directly to each circuit. This appendix records two general consequences of Definition 1.1 that are useful elsewhere.

If a joint operator is displayed as a 2×2 block matrix whose row and column are indexed by the ancilla, then the ket on the right of (1.a.15) selects an ancilla column and the bra on the left selects the matching row. Hence

$$X = \begin{pmatrix} X_{00} & X_{01} \\ X_{10} & X_{11} \end{pmatrix}_A, \qquad \mathrm{Tr}_A(X) = X_{00} + X_{11}. \tag{A.1.9}$$

More generally, for any operator Y on the retained system and any two ancilla states $|u\rangle_A, |v\rangle_A$, direct substitution gives

$$\mathrm{Tr}_A(Y \otimes |u\rangle\langle v|_A) = \sum_{a\in\{0,1\}} \langle a|u\rangle\,\langle v|a\rangle\; Y = \langle v|u\rangle\; Y. \tag{A.1.10}$$

The last equality uses $\sum_a |a\rangle\langle a| = I_A$. Thus matching record states contribute their norm, whereas orthogonal record states contribute zero. The main proof uses this mechanism concretely in (1.a.17) and (1.a.26).

A.2 Clause (b): Deferred measurements

A.2.i Reversible Boolean gates

The main proof defines the reversible replacement R_f in Question 1(b) (equation (1.b.6)) and uses its constant support when compiling the ideal circuit into S. Here I give the explicit construction for every one- or two-input Boolean gate. Retaining the inputs allows us to recover them even when several inputs have the same Boolean output.

The reversible embedding. Let $f : \{0,1\}^k \to \{0,1\}$, with $k \in \{1,2\}$. In this discussion, x denotes the inputs to this individual Boolean gate. Add one target bit z and define

$$R_f |x\rangle |z\rangle := |x\rangle |z \oplus f(x)\rangle . \tag{A.2.1}$$

Applying this map twice XORs the same value into the target twice, so $R_f^2 = I$. Thus R_f is a permutation unitary. A fresh target gives $R_f |x\rangle |0\rangle = |x\rangle |f(x)\rangle$, using $k + 1 \le 3$ qubits while retaining the inputs.

Recovering the function from its truth table. For two input bits, define coefficients from the four truth-table values by

$$\begin{aligned} &a_0 := f(0,0), \quad a_1 := f(1,0) \oplus f(0,0), \quad a_2 := f(0,1) \oplus f(0,0), \\ &a_{12} := f(1,1) \oplus f(1,0) \oplus f(0,1) \oplus f(0,0). \end{aligned} \tag{A.2.2}$$

Every coefficient belongs to $\{0,1\}$. They give the algebraic normal form

$$f(x_1,x_2) = a_0 \oplus a_1 x_1 \oplus a_2 x_2 \oplus a_{12} x_1 x_2. \tag{A.2.3}$$

Here multiplication is AND, and addition is XOR. To verify the identity, let g denote its right-hand side. Substitution at all four inputs gives

$$\begin{aligned} &g(0,0) = a_0 = f(0,0), \quad g(1,0) = a_0 \oplus a_1 = f(1,0), \quad g(0,1) = a_0 \oplus a_2 = f(0,1), \\ &g(1,1) = a_0 \oplus a_1 \oplus a_2 \oplus a_{12} = f(1,1). \end{aligned} \tag{A.2.4}$$

Agreement on the complete truth table proves $g = f$.

A constant-size circuit. Each nonzero term in (A.2.3) contributes one toggle of the target. Use X on that target when $a_0 = 1$, CNOT from x_1 when $a_1 = 1$, CNOT from x_2 when $a_2 = 1$, and Toffoli with controls x_1, x_2 when $a_{12} = 1$. The inputs remain unchanged throughout. These target toggles compose by XOR, so their final effect is

$$|x_1,x_2\rangle |z\rangle \longmapsto |x_1,x_2\rangle |z \oplus a_0 \oplus a_1 x_1 \oplus a_2 x_2 \oplus a_{12} x_1 x_2\rangle = R_f |x_1,x_2\rangle |z\rangle . \tag{A.2.5}$$

At most one X, two CNOTs, and one Toffoli are needed, all on the same three qubits. By linearity, the equality on basis states establishes the circuit's action on any superposition, including inputs entangled with other wires.

For a unary function, define $a_0 := f(0)$ and $a_1 := f(1) \oplus f(0)$. Then

$$f(x) = a_0 \oplus a_1 x, \qquad R_f |x\rangle |z\rangle = |x\rangle |z \oplus a_0 \oplus a_1 x\rangle . \tag{A.2.6}$$

Only the corresponding X and CNOT terms are needed. Thus every Boolean gate allowed in Question 1(b) has a fixed-size unitary replacement.

These are ideal auxiliary-gate counts: X, CNOT, and Toffoli are not assumed to belong to S. The main proof's compilation step (equation (1.b.13)) approximates each complete R_f using gates from S and supplies the required accuracy and gate count.

Main-text reference: Equation (1.b.6).

A.2.ii Histories, branch probabilities, and mixed inputs

The main exactness proof (Section 1.b.ii) postpones the measurements of retained records. Here I give an alternative proof that tracks the state after each instruction. I first derive the branch probability and conditional state from Part (a)'s projector rule, then use them to verify the complete joint output distribution. The final fact extends exact equality, or a uniform error bound, from pure to mixed inputs.

From the trace rule to a squared norm. Let $|\psi\rangle$ be a normalized state of the data register, and let $P_z = (|z\rangle\langle z|)_q \otimes I$ project onto outcome $z \in \{0,1\}$ on the measured qubit q, acting as the identity on all other data

wires. Part (a) uses $p_z = \mathrm{Tr}(P_z\rho)$, as in (1.a.3). For $\rho = |\psi\rangle\langle\psi|$, this gives

$$p_z = \mathrm{Tr}(P_z |\psi\rangle\langle\psi|) = \langle\psi| P_z |\psi\rangle = \langle\psi| P_z^\dagger P_z |\psi\rangle = \|P_z |\psi\rangle\|_2^2 . \tag{A.2.7}$$

The trace equality follows by summing diagonal matrix elements; the next equality uses $P_z^\dagger = P_z = P_z^2$. Thus the probability is the squared length of the component selected by the projector.[7]

If $p_z > 0$, the conditional density matrix in (1.a.5) is the outer product of the normalized projected vector:

$$\frac{P_z |\psi\rangle\langle\psi| P_z}{p_z} = |\psi_z\rangle\langle\psi_z| , \qquad |\psi_z\rangle := \frac{P_z |\psi\rangle}{\|P_z |\psi\rangle\|_2} . \tag{A.2.8}$$

Projection selects the outcome component; normalization makes that component a unit vector representing the state after the outcome is known.

The state after each instruction. Fix a normalized pure quantum input $|\psi_0\rangle$ and a classical input x. The register D holds the evolving quantum data, R the fresh measurement records, and W the fresh Boolean outputs. As in Definition 1.2, a history h is a possible list of classical values generated by the first t instructions of C, with a zero placeholder for each inactive measurement. Write $\Pr_C[h]$ for the probability that these instructions produce exactly that list. The fixed input x remains separate; together, h and x determine the later classical control decisions.

The same string h labels the basis state $|h\rangle_{\mathrm{RW}}$ in C_{ideal}. The claim compares the weight of this coherent component with the probability of the corresponding classical history in C. Omitting unchanged input qubits and unused fresh zeros, after t instructions the coherent circuit satisfies

$$|\Psi_t\rangle = \sum_h |h\rangle_{\mathrm{RW}} \left|\phi_{t,h}\right\rangle_{\mathrm{D}} , \qquad \left\|\phi_{t,h}\right\|_2^2 = \Pr_C[h] . \tag{A.2.9}$$

Whenever $\Pr_C[h] > 0$, the normalized vector $\left|\phi_{t,h}\right\rangle / \left\|\phi_{t,h}\right\|_2$ is exactly the data state of C conditioned on h. Initially the unique empty history has probability one and data state $|\psi_0\rangle$.

An unconditional unitary applies the same U to the data in both circuits. A controlled unitary reads the same control value and therefore makes the same choice between I and U. A Boolean replacement appends the deterministic value $f(h)$, with dependence on the fixed x understood, while preserving its inputs, the data state, and the branch norm. Thus these three instruction types preserve the asserted state description.

Checking a measurement instruction. If the next instruction measures qubit q unconditionally, write

$$\left|\phi_{t,h}\right\rangle = |0\rangle_q \left|\eta_{h,0}\right\rangle + |1\rangle_q \left|\eta_{h,1}\right\rangle . \tag{A.2.10}$$

The coherent circuit applies only the recorder:

$$|\Psi_t\rangle |0\rangle_r \xrightarrow{\mathrm{CNOT}_{q\to r}} \sum_h \sum_{z=0}^{1} |h, z\rangle_{\mathrm{RW}} |z\rangle_q \left|\eta_{h,z}\right\rangle . \tag{A.2.11}$$

This updates every history coherently, without measuring. The label z denotes a possible outcome of the actual measurement in C.

For a history of positive probability, the normalized data state is $\left|\phi_{t,h}\right\rangle / \left\|\phi_{t,h}\right\|_2$. Projection onto z selects exactly the data component copied by the recorder:

$$P_z\left(|0\rangle_q \left|\eta_{h,0}\right\rangle + |1\rangle_q \left|\eta_{h,1}\right\rangle\right) = |z\rangle_q \left|\eta_{h,z}\right\rangle , \qquad z \in \{0, 1\}. \tag{A.2.12}$$

The other component vanishes because $\langle z|1 - z\rangle = 0$. Applying (A.2.7) to the normalized history state therefore gives

$$\Pr_C[z \mid h] = \left\| P_z \frac{\left|\phi_{t,h}\right\rangle}{\left\|\phi_{t,h}\right\|_2} \right\|_2^2 = \frac{\left\|\eta_{h,z}\right\|_2^2}{\left\|\phi_{t,h}\right\|_2^2} . \tag{A.2.13}$$

The denominator removes the weight of reaching h, leaving the probability of the next outcome *conditioned on having reached that history*. Multiplying that weight back in gives

$$\Pr_C[h, z] = \Pr_C[h] \Pr_C[z \mid h] = \left\|\eta_{h,z}\right\|_2^2 , \qquad \left|\psi_{h,z}\right\rangle = \frac{|z\rangle_q \left|\eta_{h,z}\right\rangle}{\left\|\eta_{h,z}\right\|_2} \quad \text{if } \Pr_C[h, z] > 0. \tag{A.2.14}$$

[7] The projective-measurement rule is stated in M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Sec. 2.2.3.

The recorder retains precisely this component and tags it with $|h, z\rangle$. Its squared norm and normalized data state therefore match the probability and conditional state of the original measurement. If the component is zero, that outcome has probability zero and no conditional state is assigned to it.

For a conditional measurement, the earlier values in h, together with the fixed input x, determine its enable bit e. When $e = 1$, C performs the measurement and C_{ideal} applies the recorder just analyzed. When $e = 0$, C skips the measurement and C_{ideal} leaves the data unchanged and appends zero. The retained bit e distinguishes this skipped measurement from a performed measurement whose outcome is zero. This completes the induction over all instructions. Later gates may change the data qubits, but retained records keep distinct histories orthogonal.

Recovering the complete joint output. Write the requested output values as (u, v): u comes from final measurements of the output data qubits, while v consists of the classical results that C returns from earlier measurements or Boolean gates. In C_{ideal}, those same results are read by measuring the corresponding record/work qubits at the end. Thus v denotes the same classical output string in both circuits. Let M_u project onto u on the output data wires, acting as the identity on other data wires. Let $\mathcal{H}_v$ contain the nonzero-probability histories whose requested classical results equal v, and let t_{f} denote the final instruction.

On the span of reachable histories, the projector selecting the stored output v is

$$Q_v = \sum_{h \in \mathcal{H}_v} |h\rangle\langle h| .$$

Expanding the final coherent state from (A.2.9) makes the disappearance of cross terms explicit:

$$\begin{aligned}
\Pr[C_{\text{ideal}}(|\psi_0\rangle, x) = (u, v)] &= \left\langle \Psi_{t_{\text{f}}} \right| (Q_v \otimes M_u) \left| \Psi_{t_{\text{f}}} \right\rangle \\
&= \sum_{h,h'} \langle h'| Q_v |h\rangle \left\langle \phi_{t_{\text{f}},h'} \right| M_u \left| \phi_{t_{\text{f}},h} \right\rangle \\
&= \sum_{h \in \mathcal{H}_v} \left\langle \phi_{t_{\text{f}},h} \right| M_u \left| \phi_{t_{\text{f}},h} \right\rangle \\
&= \sum_{h \in \mathcal{H}_v} \Pr_C[h] \Pr_C[u \mid h] = \Pr[C(|\psi_0\rangle, x) = (u, v)].
\end{aligned} \tag{A.2.15}$$

The history-register entry $\langle h'| Q_v |h\rangle$ is 1 when $h' = h \in \mathcal{H}_v$ and 0 otherwise, so distinct histories contribute no cross terms. The last line uses (A.2.9) and the law of total probability for C.

If no stored bits are requested, sum over all histories; if no data bits are requested, take $M_u = I$. Requested fixed input bits already agree. Thus the complete requested output distributions coincide, with unrequested records ignored only at the end.

Fact A.1 (Extension to mixed inputs)**.** Fix a classical input and let $\eta \geq 0$. If two circuits' requested output distributions have statistical distance at most η on every pure quantum input, the same bound holds on every density-matrix input.

Proof. Write

$$\rho = \sum_i \lambda_i |\psi_i\rangle\langle\psi_i| , \qquad \lambda_i \geq 0, \qquad \sum_i \lambda_i = 1. \tag{A.2.16}$$

Let p_i, q_i be the two output distributions on $|\psi_i\rangle$. By linearity, the distributions on ρ are $\sum_i \lambda_i p_i$ and $\sum_i \lambda_i q_i$. The triangle inequality gives

$$\text{SD}\left(\sum_i \lambda_i p_i, \sum_i \lambda_i q_i \right) \leq \sum_i \lambda_i \text{SD}(p_i, q_i) \leq \eta. \tag{A.2.17}$$

■

Applying Fact A.1 with $\eta = 0$ extends (A.2.15) to every density matrix. This independently proves the main exactness claim (Section 1.b.ii) (1.b.10) for every quantum input and fixed classical input x.

A.2.iii Compilation conventions and error estimates

The compilation step in Question 1(b) (Section 1.b.iii) replaces each additional gate by a circuit over the prescribed set S. I give the fixed-block and global-phase conventions behind (1.b.13), then derive the circuit and output-distribution error bounds used in the main proof. The calculations make explicit why local approximation errors add and why the same bound controls the requested final probabilities.

Why a fixed block is enough. The auxiliary library is finite and its gates have bounded support. Choose one constant block size on which S supports universal synthesis of all these operations. For example, if a two-qubit target must be synthesized using three-qubit gates, its intended action on the third wire is the identity. The target on the full block is therefore $G \otimes I$. A shared constant-size register provides spare wires whenever they are needed.

The approximation is in operator norm on the whole block, not just on inputs whose spare wires are zero. The synthesized circuit need not return those wires exactly to zero. Nevertheless, at each occurrence its difference from the ideal $G \otimes I$ is bounded uniformly, even when the spare wires are correlated with other registers. Tensoring with the identity on the remaining wires preserves this norm bound. Consequently the same telescoping estimate as in (1.b.14) permits reuse of the spare register without an exact reset between replacements.

The determinant-one formulation. The fixed-dimensional theorem in [4, Theorem 1 and Sec. 5] is stated for $SU(d)$, where d is the dimension of the fixed block. Multiplying a unitary by a suitable scalar phase makes its determinant one. Universality up to global phase gives density of determinant-one representatives; taking their adjoints as well gives an inverse-closed finite instruction set. Each representative is the corresponding allowed gate placement multiplied by a scalar phase. Applying the theorem to these representatives and then restoring the original gates therefore gives exactly the approximation up to a global phase in (1.b.13). Its length bound remains unchanged.

Why the complete controlled gate is the target. A phase that is global before adding a control need not remain global afterward. Even if an approximation differs from U only by a phase, adding a control gives

$$\operatorname{ctrl}(e^{i\theta}U) = |0\rangle\langle 0| \otimes I + e^{i\theta}\,|1\rangle\langle 1| \otimes U \neq e^{i\theta}\operatorname{ctrl}(U) \qquad \text{in general.} \tag{A.2.18}$$

The first expression changes the relative phase between the two control branches. Compiling $\operatorname{ctrl}(U)$ directly avoids this problem: the approximation's allowed phase multiplies the complete controlled operation. Across all replacements, these phases multiply to the single unobservable overall phase used in the main error estimate.

Why the individual errors add. The main proof invokes the course's additive-error estimate. Lecture 3 proves it by induction on the evolving state vector; the following operator-product calculation gives the same bound uniformly over all inputs.[8] All operations below are extended by the identity to the full circuit registers.

For the complete circuit, list the ideal operations as $A_1, \ldots, A_K$ and their replacements as $B_1, \ldots, B_K$. At an unchanged position $B_j = A_j$; at one of the L additional-gate positions, $B_j = e^{i\theta_j}\widetilde{G}_j$ and $\|A_j - B_j\| \leq \delta$. Write $U_{\text{ideal}} = A_K \cdots A_1$ for the ideal unitary and U' for the implemented unitary before the final measurements. Adding and subtracting intermediate products gives the telescoping identity

$$U_{\text{ideal}} - e^{i\Theta}U' = \sum_{j=1}^{K} A_K \cdots A_{j+1}(A_j - B_j)B_{j-1} \cdots B_1, \qquad \Theta = \sum_{j \text{ auxiliary}} \theta_j. \tag{A.2.19}$$

Expanding each $A_j - B_j$ makes adjacent terms cancel, leaving the difference of the two full products. The operators surrounding each difference are unitary, so they preserve its norm. Taking norms and applying the triangle inequality therefore gives the sum of the individual errors. Only the L replaced positions contribute, yielding the main bound (1.b.14).

From circuit error to output probabilities. For the same complete normalized input $|\psi\rangle$, including the fixed classical input $|x\rangle$ and fresh zero qubits, define the final states with the unobservable phase absorbed into the second:

$$|\alpha\rangle = U_{\text{ideal}}\,|\psi\rangle\,, \qquad |\beta\rangle = e^{i\Theta}U'\,|\psi\rangle\,. \tag{A.2.20}$$

The operator-norm bound (1.b.14) gives

$$\||\alpha\rangle - |\beta\rangle\|_2 = \left\|(U_{\text{ideal}} - e^{i\Theta}U')\,|\psi\rangle\right\|_2 \leq L\delta. \tag{A.2.21}$$

The state-to-distribution estimate was stated as an exercise in Lecture 2, with Cauchy–Schwarz as the proof hint.[9] Here the output may include selected data, record, and work bits. Write a complete computational-basis label as (y, g), where y is the requested output and g collects every unobserved wire, including unrequested

[8] Course Lecture 3: additive accumulation of approximation errors and the choice of per-gate accuracy.

[9] Course Lecture 2: the relation between state-vector distance and the statistical distance of measurement outcomes.

record, work, padding, and input wires:

$$|\alpha\rangle = \sum_{y,g} \alpha_{y,g} \, |y, g\rangle \, , \qquad |\beta\rangle = \sum_{y,g} \beta_{y,g} \, |y, g\rangle \, . \tag{A.2.22}$$

The requested probabilities are $p(y) = \sum_g |\alpha_{y,g}|^2$ and $q(y) = \sum_g |\beta_{y,g}|^2$. Consequently,

$$\begin{aligned} \mathrm{SD}(p, q) &= \frac{1}{2} \sum_y \left| \sum_g \left(|\alpha_{y,g}|^2 - |\beta_{y,g}|^2 \right) \right| \\ &\leq \frac{1}{2} \sum_{y,g} |\alpha_{y,g} - \beta_{y,g}| (|\alpha_{y,g}| + |\beta_{y,g}|) \leq \| |\alpha\rangle - |\beta\rangle \|_2 \leq L\delta. \end{aligned} \tag{A.2.23}$$

The first inequality uses the triangle inequality and $\big||a| - |b|\big| \leq |a - b|$. The second uses Cauchy–Schwarz, since the Euclidean norm of the vector with entries $|\alpha_{y,g}| + |\beta_{y,g}|$ is at most $\|\alpha\|_2 + \|\beta\|_2 = 2$. The sum over g is ordinary marginalization: auxiliary outputs are ignored only after they have completed all their control tasks. This proves the pure-input estimate used in (1.b.15); Fact A.1 supplies its extension to mixed inputs.

Main-text reference: Section 1.b.iii.

APPENDIX B

Question 2: Local information in encoded states

These discussions expand the state-level interpretation in P2a, the basis-choice qualification in P2b, the erasure and partial-trace reasoning in Q2(a), and the Pauli-coordinate explanation and local examples in Q2(b). The complete eigenvector and Pauli-basis proofs of P2b appear in the main solution.

B.1 P2a: Coset labels after transversal Hadamard

The state interpretation in P2a (Section P2a.v) concerns a code $C = \mathrm{CSS}(C_1, C_2)$, where $C_2 \subseteq C_1$. Its computational logical basis consists of uniform coset states $|x + C_2\rangle = |C_2|^{-1/2} \sum_{c \in C_2} |x + c\rangle$, with $x \in C_1$. The main proof identifies the transformed code as $\mathrm{CSS}(C_2^\perp, C_1^\perp)$. The following formula describes where an individual coset state lies within that code.

I obtain the state-level transformation directly by expanding the Hadamard action on each computational string:

$$H^{\otimes n} |x + C_2\rangle = \frac{1}{\sqrt{2^n |C_2|}} \sum_{y \in \{0,1\}^n} (-1)^{x \cdot y} \left(\sum_{c \in C_2} (-1)^{c \cdot y} \right) |y\rangle . \tag{B.1.1}$$

The inner sum is $|C_2|$ when $y \in C_2^\perp$. Otherwise there is a $c_0 \in C_2$ with $c_0 \cdot y = 1$; pairing c with $c + c_0$ cancels the summands in pairs. Using $|C_2||C_2^\perp| = 2^n$ therefore gives

$$H^{\otimes n} |x + C_2\rangle = \frac{1}{\sqrt{|C_2^\perp|}} \sum_{y \in C_2^\perp} (-1)^{x \cdot y} |y\rangle . \tag{B.1.2}$$

The logical coset label x has not disappeared. Before Hadamard it is represented by the displaced computational strings $x + c$; afterwards it is represented by the relative phases $(-1)^{x \cdot y}$. The transformed state is generally a logical phase-basis state of the new CSS code rather than one single canonical coset-basis state. This is the state-level meaning of the exchange between the X- and Z-descriptions.

Main-text reference: Section P2a.v.

B.2 P2b: When two bases force maximal mixing

The basis-choice qualification in P2b (Section P2b.i) concerns how the second basis mixes the first. In this discussion, d denotes the Hilbert-space dimension.

This argument also isolates what the choice of bases contributes. The computational basis is not essential.

A simple sufficient condition. Suppose that, in some first basis, $\rho |a_i\rangle = p_i |a_i\rangle$, and that a second diagonalizing basis contains

$$|b\rangle = \sum_{i=1}^{d} c_i |a_i\rangle \qquad \text{with } c_i \neq 0 \text{ for every } i, \tag{B.2.1}$$

then $\rho |b\rangle = \lambda |b\rangle$ gives $c_i(p_i - \lambda) = 0$. Since every c_i is nonzero, this forces $p_i = \lambda$ for all i.

The exact condition. More generally, whenever one vector of the second basis contains nonzero components along both $|a_i\rangle$ and $|a_j\rangle$, the same eigenvector equation forces $p_i = p_j$. The two diagonality assumptions force maximal mixing exactly when these overlaps connect all first-basis directions. If they split into separate groups, each group can carry its own constant eigenvalue.

Why arbitrary two bases do not suffice. For example, applying H only to the first of two qubits mixes $\{|00\rangle, |10\rangle\}$ and $\{|01\rangle, |11\rangle\}$ separately, so $\mathrm{diag}(a, b, a, b)$ is diagonal in both bases even when $a \neq b$ (with $2a + 2b = 1$). The full Hadamard basis avoids such separate groups because $|+\rangle^{\otimes n}$ has a nonzero component along every computational string.

Main-text reference: Section P2b.i.

B.3 Q2(a): Erasure intuition and partial-trace details

I collect here the erasure motivation, the expanded partial-trace calculation, and the proof map for Question 2(a) (Section 2.a). Throughout, S is the retained subsystem of $t = d - 1$ qubits and R is its complement, as in the main setup (Section 2.a.i).

Remark B.1 (Erasure versus local information)**.** The same partition appears in two complementary questions:

Question	Physically retained	Traced out or lost
What state does the small subsystem have?	S	R
What remains if the small subsystem is erased?	R	S

Thus $\rho_S = \mathrm{Tr}_R\,|\phi\rangle\langle\phi|$ describes what is visible inside S. The erasure experiment instead loses S and asks whether R can recover the encoded state.

The standard CSS distance formula gives

$$d_{\mathrm{CSS}} = \min\left\{ \min_{x\in C_1\setminus C_2} \mathrm{wt}(x),\ \min_{z\in C_2^{\perp}\setminus C_1^{\perp}} \mathrm{wt}(z) \right\} \geq d, \tag{B.3.1}$$

where a minimum over an empty set is $+\infty$. Every word in either minimization is nonzero and lies in a classical code whose distance is at least d. By the distance/erasure theorem, erasure of the known positions S is therefore correctable because $|S| < d$. Concretely, one recovery channel $\mathcal{R}_R$, taking the remaining subsystem R as input and reconstructing the encoded state, satisfies

$$\mathcal{R}_R(\mathrm{Tr}_S\,|\psi_L\rangle\langle\psi_L|) = |\psi_L\rangle\langle\psi_L| \qquad \text{for every normalized } |\psi_L\rangle \in C. \tag{B.3.2}$$

Thus R retains everything needed to reconstruct whichever logical state was encoded. The complementary no-leakage intuition is that S should look identical for all encoded inputs, suggesting

$$\mathrm{Tr}_R\,|\psi_L\rangle\langle\psi_L| = \mathrm{Tr}_R\,|\varphi_L\rangle\langle\varphi_L| \qquad \text{for all normalized } |\psi_L\rangle, |\varphi_L\rangle \in C. \tag{B.3.3}$$

This is motivation, not the proof used in Q2(a): the direct calculation establishes the stronger claim $\rho_S = I_{2^t}/2^t$. An input-independent subsystem could still be fixed in a biased state; a maximally mixed subsystem has no fixed local bias at all.

Main-text reference: Section 2.a.i.

B.3.i Why a unitary on the discarded subsystem disappears

In Q2(a), I apply Hadamard to every qubit and then keep only S. The proof needs to identify the resulting state on S: it should be the same as first discarding R and then applying Hadamard only to S. The apparent obstacle is that the global transformation also acts on R. I first explain why that action has no effect on the reduced state, using one outer-product term; then I extend the calculation to an arbitrary joint state and recover Fact 2.1.

What partial trace keeps from one term. Let $|a\rangle, |b\rangle$ be vectors on S, and let $|r\rangle, |s\rangle$ be vectors on R. Consider the joint operator

$$A = \big(|a\rangle\langle b|\big) \otimes \big(|r\rangle\langle s|\big). \tag{B.3.4}$$

Such a term need not itself be a density matrix: off-diagonal terms are among the pieces from which a density matrix is assembled. Tracing out R leaves the operator on S multiplied by the trace of the operator on R:

$$\mathrm{Tr}_R(A) = |a\rangle\langle b|\ \mathrm{Tr}\big(|r\rangle\langle s|\big) = |a\rangle\langle b|\ \langle s|r\rangle\,. \tag{B.3.5}$$

The last equality follows by summing diagonal entries in any orthonormal basis $\{|j\rangle\}$ of R:

$$\mathrm{Tr}\big(|r\rangle\langle s|\big) = \sum_j \langle j|r\rangle\,\langle s|j\rangle = \langle s|\left(\sum_j |j\rangle\langle j|\right)|r\rangle = \langle s|r\rangle\,. \tag{B.3.6}$$

Thus the concrete rule is: *tracing out R replaces its outer product by an inner product*. The retained factor $|a\rangle\langle b|$ stays in place.

For example, keeping the first qubit and discarding the second gives

$$\begin{aligned} \mathrm{Tr}_2\big(|00\rangle\langle 11|\big) &= |0\rangle\langle 1|\ \langle 1|0\rangle = 0, \\ \mathrm{Tr}_2\big(|00\rangle\langle 10|\big) &= |0\rangle\langle 1|\ \langle 0|0\rangle = |0\rangle\langle 1|\,. \end{aligned} \tag{B.3.7}$$

Different orthogonal labels on the discarded qubit kill the first term; identical labels let the second survive. In particular, partial trace does not automatically remove every off-diagonal operator on S.

Why a unitary on R changes nothing. Now apply a unitary V_R only to the discarded subsystem. The term A becomes

$$(I_S \otimes V_R)A(I_S \otimes V_R^\dagger) = |a\rangle\langle b| \otimes |V_R r\rangle\langle V_R s|\,, \tag{B.3.8}$$

where $|V_R r\rangle := V_R\,|r\rangle$ and similarly for $|V_R s\rangle$. Applying the same partial-trace rule gives

$$\begin{aligned}\mathrm{Tr}_R\Big[(I_S \otimes V_R)A(I_S \otimes V_R^\dagger)\Big] &= |a\rangle\langle b|\ \langle V_R s|V_R r\rangle \\ &= |a\rangle\langle b|\ \langle s|\,V_R^\dagger V_R\,|r\rangle \\ &= |a\rangle\langle b|\ \langle s|r\rangle = \mathrm{Tr}_R(A).\end{aligned} \tag{B.3.9}$$

The decisive step is $V_R^\dagger V_R = I_R$: *a unitary preserves exactly the inner product that survives the partial trace.*

For a discarded qubit, Hadamard changes $|0\rangle$ into $|+\rangle$ and $|1\rangle$ into $|-\rangle$. In the two examples above, the scalar factors therefore become $\langle -|+\rangle = 0$ and $\langle +|+\rangle = 1$, respectively. The same term vanishes and the same term survives. The global operator can change; its reduced operator on the retained subsystem is unchanged.

Why this also covers entangled states. The calculation above does not assume that the joint state is a product. Choose orthonormal bases $\{|a\rangle\}$ of S and $\{|r\rangle\}$ of R. Every joint density operator has an expansion

$$\rho_{SR} = \sum_{a,b,r,s} c_{abrs}\,|a\rangle\langle b| \otimes |r\rangle\langle s|\,, \qquad c_{abrs} := (\langle a| \otimes \langle r|)\rho_{SR}(|b\rangle \otimes |s\rangle). \tag{B.3.10}$$

This is simply its matrix-entry expansion in a product basis. The coefficients need not factor, and the sum is not asserted to be a mixture of product states. It includes entangled states and their cross terms. Conjugation and partial trace are both linear, so (B.3.9) applies to every summand and hence to the whole density operator.

If a unitary U_S is also applied to the retained subsystem, its effect on each retained factor is $|a\rangle\langle b| \mapsto U_S\,|a\rangle\langle b|\,U_S^\dagger$. That operator remains after tracing out R. Summing the terms therefore gives the full identity

$$\boxed{\mathrm{Tr}_R\Big[(U_S \otimes V_R)\rho_{SR}(U_S^\dagger \otimes V_R^\dagger)\Big] = U_S\,\mathrm{Tr}_R(\rho_{SR})U_S^\dagger.} \tag{B.3.11}$$

The equivalent basis-sum derivation. I retain the direct derivation of Fact 2.1 to connect this term-by-term mechanism to the compact formula used in the main proof. Let ρ_{SR} be a density operator, and let U_S and V_R be unitaries on the retained and discarded subsystems, respectively. Choose any orthonormal basis $\{|j\rangle_R\}$ of R. The definition of partial trace gives

$$\begin{aligned}\mathrm{Tr}_R\Big[(U_S \otimes V_R)\rho_{SR}(U_S^\dagger \otimes V_R^\dagger)\Big] &= \sum_j (I_S \otimes \langle j|_R)(U_S \otimes V_R)\rho_{SR}(U_S^\dagger \otimes V_R^\dagger)(I_S \otimes |j\rangle_R) \\ &= U_S\left[\sum_j (I_S \otimes \langle j|_R\,V_R)\rho_{SR}(I_S \otimes V_R^\dagger\,|j\rangle_R)\right]U_S^\dagger.\end{aligned} \tag{B.3.12}$$

The last equality pulls out the operators on S. Put $\big|\widetilde{j}\big\rangle_R := V_R^\dagger\,|j\rangle_R$. Unitarity makes $\{\big|\widetilde{j}\big\rangle_R\}$ another orthonormal basis, with $\langle j|_R\,V_R = \big\langle\widetilde{j}\big|_R$. The bracket is therefore

$$\sum_j (I_S \otimes \big\langle\widetilde{j}\big|_R)\rho_{SR}(I_S \otimes \big|\widetilde{j}\big\rangle_R) = \mathrm{Tr}_R(\rho_{SR}), \tag{B.3.13}$$

which proves the fact. A unitary on the retained subsystem remains as a conjugation; a unitary acting solely on the subsystem that is traced out amounts only to a change of basis in the summation.

Return to the Hadamard step in Q2(a). Write $H_S = H^{\otimes t}$ and $H_R = H^{\otimes(n-t)}$, so that $H^{\otimes n} = H_S \otimes H_R$. Starting from the global encoded state $|\phi\rangle$, define $\rho_S = \mathrm{Tr}_R\,|\phi\rangle\langle\phi|$ and $|\phi_H\rangle = (H_S \otimes H_R)\,|\phi\rangle$. Since Hadamard is its own adjoint, the identity just proved gives

$$\begin{aligned}\widetilde{\rho}_S := \mathrm{Tr}_R\,|\phi_H\rangle\langle\phi_H| &= \mathrm{Tr}_R[(H_S \otimes H_R)\,|\phi\rangle\langle\phi|\,(H_S \otimes H_R)] \\ &= H_S\,\mathrm{Tr}_R(|\phi\rangle\langle\phi|)H_S = H_S\rho_S H_S.\end{aligned} \tag{B.3.14}$$

Thus applying Hadamard globally and then discarding R gives the same state on S as first discarding R and then applying Hadamard to S. The H_R factors disappear because they preserve the discarded inner products; the H_S

factors remain because they act on the retained operators.

This is the precise bridge needed in Q2(a). The distance argument applied to the transformed CSS code shows that $\widetilde{\rho}_S$ is computational-basis diagonal. Equation (B.3.14) identifies it as $H_S \rho_S H_S$. Together with the already established diagonality of ρ_S, these are exactly the two hypotheses of P2b, which yields $\rho_S = I_{2^t}/2^t$.

Main-text reference: Fact 2.1.

B.3.ii Proof map: where every hypothesis enters

Input	Exact use
$\lvert S\rvert = t < d$ and $\mathrm{dist}(C_1) \geq d$	Exclude a surviving computational-basis off-diagonal term: (2.a.23).
P2a and $\mathrm{dist}(C_2^{\perp}) \geq d$	Repeat the same argument after transversal Hadamard: (2.a.26).
Partial-trace covariance	Identify the transformed reduced state as $H^{\otimes t}\rho_S H^{\otimes t}$, using (2.a.27).
P2b	Upgrade diagonality in both complementary bases to maximal mixing: (2.a.33).
Coordinate permutation	Extend the first-t-positions proof to an arbitrary small subsystem: the coordinate-reduction argument (Section 2.a.ii).

The distance of C_1 removes coherences visible in the computational description. P2a moves the argument to the complementary CSS side, where the distance of $C_2^{\perp}$ removes the Hadamard-basis coherences. P2b then leaves only the normalized identity.

Main-text reference: Equation (2.a.33).

B.4 Q2(b): Pauli coordinates and local information

I expand here three steps in Q2(b): why Pauli expectations determine a matrix, how a local expectation is evaluated on the full encoded state, and which stabilizers fit inside the two pairs in the Shor example. Throughout, T is the selected set of $t < d$ physical qubits, $W = \overline{T}$, and $\rho_T^{\phi} = \mathrm{Tr}_W \lvert\phi\rangle\langle\phi\rvert$ for a normalized codeword $\lvert\phi\rangle \in C$.

B.4.i Pauli coordinates and measurement meaning

The main proof compares ρ_T^{ϕ} and ρ_T^{ψ} by comparing their traces against Pauli matrices. I develop here the two connections behind that step: why these numbers determine the whole matrix, and why they are expectations of physical measurement outcomes. The first connection is linear algebra; the second requires the Born rule. Keeping their derivations separate makes clear what each contributes to Q2(b).

Matrices and their coordinates. A matrix can be expressed as a linear combination of other matrices, just as a vector can be expressed in a vector basis. For example,

$$|0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \frac{1}{2}\left[\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\right] = \frac{1}{2}(I + Z). \tag{B.4.1}$$

The left side gives the matrix as a state projector; the right side gives the same operator in terms of I and Z, with coefficient $1/2$ for each. These are two descriptions of one matrix.

For a general t-qubit matrix, the proposed coordinate family is

$$\mathcal{B}_t = \{I, X, Y, Z\}^{\otimes t}, \qquad P = P_1 \otimes \cdots \otimes P_t, \qquad P_j \in \{I, X, Y, Z\}. \tag{B.4.2}$$

The notation $\{I, X, Y, Z\}^{\otimes t}$ means that each of the t positions independently receives one of the four matrices

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{B.4.3}$$

For $t = 1$, the family consists of these four 2×2 matrices. For $t = 2$, there are $4 \cdot 4 = 16$ choices: the first and second factors can be chosen separately. For example, $X \otimes Z$ means that X acts on the first qubit and Z on the second. The tensor product is itself one 4×4 matrix. Replacing each entry of X by that entry times the block Z

gives

$$X \otimes Z = \begin{pmatrix} 0Z & 1Z \\ 1Z & 0Z \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}. \tag{B.4.4}$$

The rows and columns use the order $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. Its second column is the image of $|01\rangle$, namely $-|11\rangle$. The same result follows by applying the two factors separately:

$$(X \otimes Z)|01\rangle = (X|0\rangle) \otimes (Z|1\rangle) = |1\rangle \otimes (-|1\rangle) = -|11\rangle. \tag{B.4.5}$$

Thus a Pauli string describes one operator through its actions on separate qubits. In general, t factors give a $2^t \times 2^t$ matrix and 4^t possible strings. To use these matrices as coordinates, we must establish that they form a basis and then find the coefficients.

Why the Pauli strings form a basis. The needed property is orthogonality under the trace inner product. I first calculate the one-qubit products that make it work. Equal factors give the identity. For example, ordinary matrix multiplication gives

$$\begin{aligned} X^2 &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I, \\ Y^2 &= \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} (-i)i & 0 \\ 0 & i(-i) \end{pmatrix} = I. \end{aligned} \tag{B.4.6}$$

Likewise, $Z^2 = \mathrm{diag}(1^2, (-1)^2) = I$, and $I^2 = I$. In every case the trace of the product is $\mathrm{Tr}(I) = 1 + 1 = 2$.

For distinct factors there are two possibilities. If one is I, the product is the other matrix. The displayed diagonals give

$$\mathrm{Tr}(IX) = 0 + 0 = 0, \qquad \mathrm{Tr}(IY) = 0 + 0 = 0, \qquad \mathrm{Tr}(IZ) = 1 + (-1) = 0. \tag{B.4.7}$$

Putting I on the right gives the same products. If neither factor is I, their product is another nonidentity Pauli times a phase. For example,

$$XY = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = iZ, \qquad \mathrm{Tr}(XY) = i + (-i) = 0. \tag{B.4.8}$$

The same multiplication for the other ordered pairs gives

$$\begin{aligned} XY &= iZ, \quad YZ = iX, \quad ZX = iY, \\ YX &= -iZ, \quad ZY = -iX, \quad XZ = -iY. \end{aligned} \tag{B.4.9}$$

The phase and the order matter for the product, but each result still has trace zero. These calculations account for every pair of one-qubit factors and establish

$$\mathrm{Tr}(P_j Q_j) = \begin{cases} 2, & P_j = Q_j, \\ 0, & P_j \neq Q_j. \end{cases} \tag{B.4.10}$$

For several qubits, two tensor-product rules extend this calculation. Multiplication acts on matching positions: $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$. Indeed, on a product vector $|u\rangle \otimes |v\rangle$, the successive actions give $A(C|u\rangle) \otimes B(D|v\rangle)$, which is exactly the action of $(AC) \otimes (BD)$. Product basis vectors span the full space, so this verifies the operator identity.

Trace also multiplies over tensor products. To see this directly, let $R = (r_{ab})_{a,b=0}^{1}$ be a 2×2 matrix and let S be any square matrix. Only the diagonal blocks of $R \otimes S$ contribute to its trace, so

$$\begin{aligned} \mathrm{Tr}(R \otimes S) &= \mathrm{Tr}\begin{pmatrix} r_{00}S & r_{01}S \\ r_{10}S & r_{11}S \end{pmatrix} \\ &= r_{00}\,\mathrm{Tr}(S) + r_{11}\,\mathrm{Tr}(S) = (r_{00} + r_{11})\,\mathrm{Tr}(S) = \mathrm{Tr}(R)\,\mathrm{Tr}(S). \end{aligned} \tag{B.4.11}$$

Repeating this rule handles any number of qubit factors.

For a concrete comparison, take $P = X \otimes Z$. Equal strings give

$$P^2 = (X \otimes Z)(X \otimes Z) = X^2 \otimes Z^2 = I \otimes I = I_4, \qquad \mathrm{Tr}(P^2) = 4. \tag{B.4.12}$$

Now change only the first factor, taking $Q = Y \otimes Z$. Then

$$\begin{aligned} PQ &= (X \otimes Z)(Y \otimes Z) = (XY) \otimes Z^2 = iZ \otimes I, \\ \operatorname{Tr}(PQ) &= \operatorname{Tr}(iZ)\operatorname{Tr}(I) = 0 \cdot 2 = 0. \end{aligned} \tag{B.4.13}$$

Equivalently, $PQ = i\,\mathrm{diag}(1, 1, -1, -1)$, so its trace is $i(1 + 1 - 1 - 1) = 0$. The product is a nonzero matrix; it is its trace that vanishes. One differing position already suffices.

For arbitrary strings $P = \bigotimes_{j=1}^{t} P_j$ and $Q = \bigotimes_{j=1}^{t} Q_j$, the same rules give

$$\operatorname{Tr}(PQ) = \operatorname{Tr}\left(\bigotimes_{j=1}^{t} P_j Q_j\right) = \prod_{j=1}^{t} \operatorname{Tr}(P_j Q_j) = 2^t \delta_{P,Q}. \tag{B.4.14}$$

If $P = Q$, every factor contributes 2. If $P \neq Q$, at least one position differs, and its zero trace makes the entire product zero. Since $P^\dagger = P$, this is orthogonality for the matrix inner product $\langle P, Q\rangle = \operatorname{Tr}(P^\dagger Q)$.

To prove linear independence explicitly, suppose $\sum_{P\in\mathcal{B}_t} c_P P = 0$. Fix one string Q, multiply the equality by Q, and take the trace:

$$0 = \operatorname{Tr}\left[\left(\sum_{P\in\mathcal{B}_t} c_P P\right) Q\right] = \sum_{P\in\mathcal{B}_t} c_P \operatorname{Tr}(PQ) = c_Q \operatorname{Tr}(Q^2) = 2^t c_Q. \tag{B.4.15}$$

All terms except $P = Q$ vanish by orthogonality, so $c_Q = 0$. The choice of Q was arbitrary; hence every coefficient is zero. There are 4^t strings, from four choices at each of t positions. The complex vector space of $2^t \times 2^t$ matrices also has dimension $(2^t)^2 = 4^t$, one coordinate for each matrix entry. The 4^t independent Pauli strings therefore form a basis of the whole space.

Extracting one coefficient. The basis conclusion gives an expansion for any matrix A:

$$A = \sum_{P\in\mathcal{B}_t} c_P P. \tag{B.4.16}$$

The coefficients exist, but their values have not yet been determined. Fix a particular $Q \in \mathcal{B}_t$. Multiplying by Q and using linearity of trace gives

$$\operatorname{Tr}(AQ) = \operatorname{Tr}\left(\sum_{P\in\mathcal{B}_t} c_P PQ\right) = \sum_{P\in\mathcal{B}_t} c_P \operatorname{Tr}(PQ). \tag{B.4.17}$$

Every term with $P \neq Q$ is zero. The one surviving term is $c_Q \operatorname{Tr}(Q^2)$, and $Q^2 = I^{\otimes t}$ has trace 2^t. Therefore

$$\operatorname{Tr}(AQ) = c_Q \operatorname{Tr}(Q^2) = 2^t c_Q, \qquad c_Q = 2^{-t} \operatorname{Tr}(AQ). \tag{B.4.18}$$

The factor 2^{-t} divides out the trace of the surviving identity matrix. Applying this calculation to every Q and substituting the coefficients into the expansion yields

$$A = 2^{-t} \sum_{P\in\mathcal{B}_t} \operatorname{Tr}(AP) P. \tag{B.4.19}$$

Thus the complete list of numbers $\operatorname{Tr}(AP)$ determines A. This establishes the algebraic part of Fact 2.2 without any measurement assumptions.

An observable and its measurement branches. I now turn to the physical meaning of these traces when the matrix being expanded is a density matrix. The new ingredient is a projective measurement. A Hermitian observable specifies orthogonal eigenspaces and a real numerical outcome for each eigenspace. A simple one-qubit example is

$$A = \begin{pmatrix} 5 & 0 \\ 0 & -2 \end{pmatrix}, \qquad A\,|0\rangle = 5\,|0\rangle\,, \qquad A\,|1\rangle = -2\,|1\rangle\,. \tag{B.4.20}$$

Here A denotes the observable. Measuring it in the normalized state $|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$ means measuring in the computational basis and reporting 5 for the $|0\rangle$ branch or -2 for the $|1\rangle$ branch. The familiar Born rule gives the probabilities $|\alpha|^2$ and $|\beta|^2$, respectively.

The projector for an outcome extracts the component in that outcome's eigenspace. In this example,

$$\Pi_5 = |0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad \Pi_{-2} = |1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}. \tag{B.4.21}$$

Their actions are explicit:

$$\begin{aligned}\Pi_5 |\psi\rangle &= |0\rangle \left(\alpha \langle 0|0\rangle + \beta \langle 0|1\rangle\right) = \alpha |0\rangle , \\ \Pi_{-2} |\psi\rangle &= |1\rangle \left(\alpha \langle 1|0\rangle + \beta \langle 1|1\rangle\right) = \beta |1\rangle .\end{aligned} \tag{B.4.22}$$

The two extracted vectors need not be normalized: their squared norms are precisely the Born probabilities. Conditional on an outcome with nonzero probability, the post-measurement state is the corresponding extracted vector divided by its norm. Here those states are $|0\rangle$ and $|1\rangle$, up to global phase.

This distinguishes a physical measurement from multiplication by the observable. Algebraically,

$$A = 5\Pi_5 - 2\Pi_{-2}, \qquad A |\psi\rangle = 5\alpha |0\rangle - 2\beta |1\rangle . \tag{B.4.23}$$

The latter vector is generally unnormalized and is not the conditional state produced by measuring A. The matrix action weights both components by their eigenvalues; measurement selects an outcome with its Born probability. More generally, the spectral expression

$$A = \sum_a a\Pi_a \tag{B.4.24}$$

packages the eigenspace projectors Π_a together with their reported values a. Changing the reported values while keeping the same distinct outcome projectors changes the numerical average, but not the branch probabilities or conditional states.

From the Born rule to projector probabilities. For the 5 outcome above, the Born rule and (B.4.22) give

$$p(5) = |\langle 0|\psi\rangle|^2 = |\alpha|^2 = \|\alpha |0\rangle\|^2 = \|\Pi_5 |\psi\rangle\|^2. \tag{B.4.25}$$

Thus the projector formula expresses the same probability as the ordinary squared-amplitude rule. For a general outcome a, the eigenspace may have an orthonormal basis $\{|v_{a,r}\rangle\}_r$ containing more than one vector. Its projector and the extracted component are

$$\Pi_a = \sum_r |v_{a,r}\rangle\langle v_{a,r}|, \qquad \Pi_a |\psi\rangle = \sum_r \langle v_{a,r}|\psi\rangle |v_{a,r}\rangle . \tag{B.4.26}$$

The outcome a groups all these basis alternatives. Their Born probabilities add, and orthonormality makes the squared norm of the extracted component the same sum:

$$p(a) = \sum_r |\langle v_{a,r}|\psi\rangle|^2 = \|\Pi_a |\psi\rangle\|^2. \tag{B.4.27}$$

Only now do the norm identity and the projector identities $\Pi_a^\dagger = \Pi_a$, $\Pi_a^2 = \Pi_a$ give

$$\begin{aligned}p(a) &= (\Pi_a |\psi\rangle)^\dagger (\Pi_a |\psi\rangle) \\ &= \langle\psi| \Pi_a^\dagger \Pi_a |\psi\rangle = \langle\psi| \Pi_a |\psi\rangle .\end{aligned} \tag{B.4.28}$$

The Born rule supplies the probability interpretation; the subsequent equalities rewrite that probability algebraically.

Density matrices, traces, and ordinary expectations. For a pure state, $\rho = |\psi\rangle\langle\psi|$. To express the last probability using ρ, expand the trace in an orthonormal basis $\{|j\rangle\}$:

$$\begin{aligned}\mathrm{Tr}(\rho\Pi_a) &= \sum_j \langle j| |\psi\rangle\langle\psi| \Pi_a |j\rangle = \sum_j \langle j|\psi\rangle \langle\psi| \Pi_a |j\rangle \\ &= \langle\psi| \Pi_a \left(\sum_j |j\rangle\langle j|\right) |\psi\rangle = \langle\psi| \Pi_a |\psi\rangle = p(a).\end{aligned} \tag{B.4.29}$$

The penultimate step uses completeness, $\sum_j |j\rangle\langle j| = I$. This is why a probability can be written as a trace against a projector.

For a mixed state $\rho = \sum_\ell r_\ell |\psi_\ell\rangle\langle\psi_\ell|$, where $r_\ell \geq 0$ and $\sum_\ell r_\ell = 1$, the same formula follows by averaging the pure-state probabilities. In an ensemble realizing this density matrix, $|\psi_\ell\rangle$ is prepared with probability r_ℓ. The

law of total probability and linearity of trace give

$$\begin{aligned} p(a) &= \sum_{\ell} r_\ell \, \langle\psi_\ell| \, \Pi_a \, |\psi_\ell\rangle \\ &= \sum_{\ell} r_\ell \, \mathrm{Tr}(|\psi_\ell\rangle\langle\psi_\ell| \, \Pi_a) = \mathrm{Tr}\left[\left(\sum_{\ell} r_\ell \, |\psi_\ell\rangle\langle\psi_\ell|\right) \Pi_a\right] = \mathrm{Tr}(\rho\Pi_a). \end{aligned} \tag{B.4.30}$$

The final expression depends only on ρ, so the choice of its ensemble decomposition does not affect the answer. It applies in particular to a reduced density matrix, which can be mixed even when the full encoded state is pure.

An expectation is now an ordinary weighted average of the numbers reported by the measurement. Writing $\mathbb{E}_\rho[A]$ for this average and substituting the probability formula gives

$$\begin{aligned} \mathbb{E}_\rho[A] &= \sum_a a \, p(a) = \sum_a a \, \mathrm{Tr}(\rho\Pi_a) \\ &= \mathrm{Tr}\left(\rho \sum_a a\Pi_a\right) = \mathrm{Tr}(\rho A). \end{aligned} \tag{B.4.31}$$

The last step is the spectral expression (B.4.24). For the concrete example, this average is $5|\alpha|^2 - 2|\beta|^2$. Thus $\mathrm{Tr}(\rho\Pi_a)$ is the probability of one outcome, while $\mathrm{Tr}(\rho A)$ averages all the numerical outcomes.

Specializing the measurement to a Pauli. For $P \in \mathcal{B}_t$, Hermiticity gives an orthonormal eigenbasis, and $P^2 = I$ forces each eigenvalue λ to satisfy $\lambda^2 = 1$. The outcomes therefore lie in $\{+1, -1\}$. The projectors can now be derived by checking their action on an eigenvector $|v_\lambda\rangle$:

$$\frac{I+P}{2} \, |v_\lambda\rangle = \frac{1+\lambda}{2} \, |v_\lambda\rangle \, , \qquad \frac{I-P}{2} \, |v_\lambda\rangle = \frac{1-\lambda}{2} \, |v_\lambda\rangle \, . \tag{B.4.32}$$

The first operator keeps the $+1$ component and annihilates the -1 component; the second does the reverse. Since the eigenvectors span the space, these are exactly the eigenspace projectors:

$$\Pi_+ = \frac{I+P}{2}, \qquad \Pi_- = \frac{I-P}{2}, \qquad \Pi_+ + \Pi_- = I, \qquad P = \Pi_+ - \Pi_- . \tag{B.4.33}$$

Using the derived Born formula and $\mathrm{Tr}(\rho) = 1$ gives

$$p_\pm = \mathrm{Tr}(\rho\Pi_\pm) = \frac{\mathrm{Tr}(\rho) \pm \mathrm{Tr}(\rho P)}{2} = \frac{1 \pm \mathrm{Tr}(\rho P)}{2}. \tag{B.4.34}$$

The same two probabilities yield

$$p_+ + p_- = 1, \qquad \mathbb{E}_\rho[P] = (+1)p_+ + (-1)p_- = p_+ - p_- = \mathrm{Tr}(\rho P). \tag{B.4.35}$$

A zero expectation therefore means $p_+ = p_- = 1/2$. Each individual measurement still reports $+1$ or -1; neither the Pauli operator nor an individual outcome is zero. For $P = I$, the minus eigenspace is absent: $\Pi_- = 0$, $\Pi_+ = I$, and only $+1$ can occur.

Returning to the reduced-state comparison. The algebraic and measurement routes now meet. Applying (B.4.19) to the reduced state gives

$$\rho_T^\phi = 2^{-t} \sum_{P \in \mathcal{B}_t} a_P^\phi P, \qquad a_P^\phi = \mathrm{Tr}(\rho_T^\phi P) = p_+^\phi - p_-^\phi. \tag{B.4.36}$$

For each fixed P, $p_\pm^\phi$ are its outcome probabilities on ρ_T^ϕ. The number a_P^ϕ is the expectation of that measurement; the coefficient multiplying the matrix P is $2^{-t} a_P^\phi$. In the opening example $\rho = |0\rangle\langle 0| = (I+Z)/2$, measuring Z reports $+1$ with certainty, so $\mathrm{Tr}(\rho Z) = 1$, whereas the coefficient of Z is $1/2$.

Finally, subtracting the expansions for two normalized codewords gives

$$\rho_T^\phi - \rho_T^\psi = 2^{-t} \sum_{P \in \mathcal{B}_t} (a_P^\phi - a_P^\psi) P. \tag{B.4.37}$$

If every Pauli expectation agrees, each coefficient on the right is zero, so the two reduced matrices are equal. Conversely, equal matrices have equal traces against each P. This is the reason Q2(b) can prove a matrix equality by fixing the scalar expectations one at a time. The next step is to evaluate each of these local expectations on the full encoded state; Section B.4.ii develops that bridge.

Main-text reference: Section 2.b.ii.

B.4.ii From a local observable to the full state

Section 2.b.iii embeds a Pauli on the selected qubits into the original physical ordering. For a concrete example, if $n \geq 4$, $T = \{2, 4\}$, and $P = X \otimes Z$, then

$$\widehat{P} = I \otimes X \otimes I \otimes Z \otimes I^{\otimes(n-4)}. \tag{B.4.38}$$

The first local factor acts on physical qubit 2, the second on qubit 4, and every other factor is identity. This embeds the observable without moving any physical qubits.

Temporarily grouping the factors as $T \otimes W$ is only bookkeeping. Applying the basis-sum rule of Definition 1.1 successively to the qubits of W gives, for any operator M and the resulting orthonormal product basis $\{|w\rangle\}$ of W,

$$\mathrm{Tr}_W M = \sum_w (I_T \otimes \langle w|) M (I_T \otimes |w\rangle). \tag{B.4.39}$$

Using also an orthonormal basis $\{|i\rangle\}$ of T, the expectation calculation is

$$\begin{aligned} \mathrm{Tr}_T[(\mathrm{Tr}_W M)P] &= \sum_{i,w} \langle i| \, (I_T \otimes \langle w|) M (I_T \otimes |w\rangle) P \, |i\rangle = \sum_{i,w} \langle i, w| \, M(P \otimes I_W) \, |i, w\rangle \\ &= \mathrm{Tr}_{TW}[M(P \otimes I_W)] \, . \end{aligned} \tag{B.4.40}$$

In the original physical ordering, $P \otimes I_W$ represents $\widehat{P}$. Taking $M = |\phi\rangle\langle\phi|$ proves the local-to-global identity (2.b.8). Its last step follows from the rank-one trace identity, which can itself be checked in any orthonormal basis $\{|i\rangle\}$:

$$\mathrm{Tr}(|u\rangle\langle v| \, A) = \sum_i \langle i|u\rangle \, \langle v| \, A \, |i\rangle = \langle v| \, A \left(\sum_i |i\rangle\langle i| \right) |u\rangle = \langle v| \, A \, |u\rangle \, . \tag{B.4.41}$$

Discarding W and measuring P on T therefore has the same expectation as measuring $\widehat{P}$ on the full state. The identity factors ensure that $\widehat{P}$ is nonidentity only inside T, so its weight is at most t; this is why the main proof can apply the distance bound to it.

Main-text reference: Section 2.b.iii.

B.4.iii Local stabilizers in the Shor example

The example in Section 2.b.v compares $T = \{1, 2\}$ with $T' = \{1, 4\}$ in the nine-qubit Shor code. Equation (2.b.17) says that the reduced state is determined by $S[T]$, the stabilizers supported entirely inside the selected set. The following enumeration checks that none are missed.

Within the first block, the two Z-type generators give exactly

$$\langle Z_1Z_2, Z_2Z_3 \rangle = \{I, Z_1Z_2, Z_2Z_3, Z_1Z_3\}, \qquad (Z_1Z_2)(Z_2Z_3) = Z_1Z_3. \tag{B.4.42}$$

Each member has an even number of Z factors in that block. The same description holds independently in blocks $\{4, 5, 6\}$ and $\{7, 8, 9\}$, so the complete Z subgroup consists precisely of the strings with even Z parity in each block.

Let G_1, G_2 be the two X-type generators. Their three nonidentity products are

$$G_1 = X_1X_2X_3X_4X_5X_6, \quad G_2 = X_4X_5X_6X_7X_8X_9, \quad G_1G_2 = X_1X_2X_3X_7X_8X_9. \tag{B.4.43}$$

Every such product occupies six positions. Multiplication by any Z-type stabilizer turns an overlapping X into Y up to phase and cannot remove its support. Consequently, every stabilizer with a nonidentity X component has weight at least six and cannot fit inside either selected pair.

It remains to inspect the Z subgroup. For $T = \{1, 2\}$, the first-block list leaves only I and Z_1Z_2; the other blocks must be identity. For $T' = \{1, 4\}$, at most one position is available in each of the first two blocks. Even parity forces both positions to be identity, and no position is available in the third block. Hence

$$S[\{1, 2\}] = \{I, Z_1Z_2\}, \qquad S[\{1, 4\}] = \{I\}. \tag{B.4.44}$$

Substitution into (2.b.17) gives the two matrices in the main example: a fixed even-parity state on $\{1, 2\}$ and the maximally mixed state on $\{1, 4\}$. Their difference records a constraint of the code, while each is independent of the logical input.

Main-text reference: Section 2.b.v.

APPENDIX C

Question 4: Discrete logarithms

These explorations expand the finite-precision and implementation arguments in Question 4. They retain the detailed circuit mechanisms and arithmetic invariants behind the contracts used in the main solution.

C.1 Phase Estimation Circuit Primer

From phase kickback to one measured finite-grid label. This primer supplements the finite-precision derivation in Question 4 (Section 4.II.i) with a circuit diagram, a grid-wave interpretation, and the circular nearest-label convention for generic phases.

This note keeps three moments separate: what the controlled powers encode, what the decoder produces, and what measurement finally selects. Its central distinction is that the PEA circuit is unitary. When the true phase is off the finite measurement grid, the circuit does not randomly choose a label; before measurement it prepares a coherent superposition

$$\sum_{z=0}^{M-1} \alpha_\theta(z)\,|z\rangle\,. \tag{C.1.1}$$

Only the subsequent measurement selects an observed label $\widetilde{z}$, with probability $|\alpha_\theta(\widetilde{z})|^2$.

C.1.i The stable setup

I begin with one fixed target eigenstate:

$$U\,|\psi\rangle = \lambda\,|\psi\rangle\,, \qquad \lambda = e^{2\pi i\theta}, \qquad 0 \le \theta < 1. \tag{C.1.2}$$

The number θ is the eigenphase measured in turns: one full turn is 1, or 2π radians. The target register T contains $|\psi\rangle$. The phase register P has m qubits, so

$$M = 2^m, \qquad z \in \{0, 1, \ldots, M-1\}, \qquad z/M \text{ is the phase represented by } z. \tag{C.1.3}$$

The familiar idealized contract

$$|0^m\rangle_P\,|\psi\rangle_T \xrightarrow{\mathrm{PEA}(U)} |\theta\rangle_P\,|\psi\rangle_T \tag{C.1.4}$$

is useful shorthand, but $|\theta\rangle$ is not generally a literal computational-basis state. The rest of this note replaces that shorthand by the actual finite-register evolution.

C.1.ii The circuit and its three stages

For $m = 3$, the conceptual circuit is

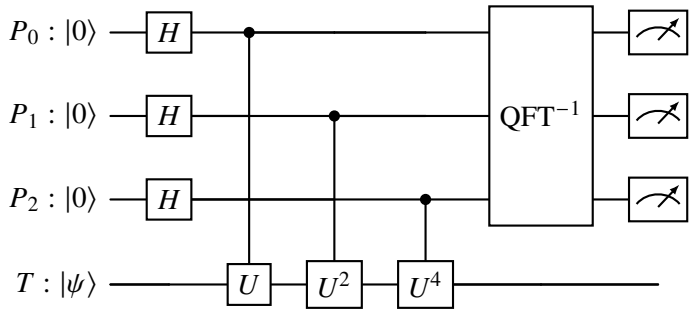


Figure C.1. Three-bit phase estimation. The controlled powers encode the phase wave, the inverse Fourier circuit decodes it into amplitudes on the eight available labels, and measurement returns one label.

The three stages are therefore

$$\text{create branches} \longrightarrow \text{encode phase} \longrightarrow \text{decode and measure}. \tag{C.1.5}$$

Stage I: create the branch labels Hadamards on the phase register give

$$|0^m\rangle_P |\psi\rangle_T \longmapsto \frac{1}{\sqrt{M}} \sum_{j=0}^{M-1} |j\rangle_P |\psi\rangle_T . \tag{C.1.6}$$

Every label j is now one coherent branch. Nothing has been measured. For $m = 3$, writing $j = b_0 + 2b_1 + 4b_2$, the three controlled gates act on branch j as

$$U^{b_0} U^{2b_1} U^{4b_2} = U^j . \tag{C.1.7}$$

The same binary-weight argument gives U^j for arbitrary m.

Stage II: write the phase wave Because the target is an eigenstate,

$$U^j |\psi\rangle = \lambda^j |\psi\rangle = e^{2\pi i j\theta} |\psi\rangle . \tag{C.1.8}$$

Hence the controlled powers transform the joint state into

$$\frac{1}{\sqrt{M}} \sum_{j=0}^{M-1} |j\rangle_P |\psi\rangle_T \longmapsto \left(\frac{1}{\sqrt{M}} \sum_{j=0}^{M-1} e^{2\pi i j\theta} |j\rangle_P \right) |\psi\rangle_T . \tag{C.1.9}$$

The target factors out unchanged. The information about θ now resides in the relative phases across the j-branches. I denote this phase wave by

$$|\Phi_\theta\rangle := \frac{1}{\sqrt{M}} \sum_{j=0}^{M-1} e^{2\pi i j\theta} |j\rangle . \tag{C.1.10}$$

Stage III: compare with the finite grid For every available label z, define the grid wave

$$|w_z\rangle := \frac{1}{\sqrt{M}} \sum_{j=0}^{M-1} e^{2\pi i j z/M} |j\rangle . \tag{C.1.11}$$

The M states $|w_0\rangle, \ldots, |w_{M-1}\rangle$ form an orthonormal basis, and the decoder satisfies $\mathrm{QFT}^{-1} |w_z\rangle = |z\rangle$. The component of the acquired wave along candidate z is

$$\alpha_\theta(z) := \langle w_z \mid \Phi_\theta\rangle = \frac{1}{M} \sum_{j=0}^{M-1} e^{-2\pi i j z/M} e^{2\pi i j\theta} = \frac{1}{M} \sum_{j=0}^{M-1} e^{2\pi i j(\theta - z/M)} . \tag{C.1.12}$$

Consequently,

$$|\Phi_\theta\rangle = \sum_{z=0}^{M-1} \alpha_\theta(z) |w_z\rangle \quad \Longrightarrow \quad \mathrm{QFT}^{-1} |\Phi_\theta\rangle = \sum_{z=0}^{M-1} \alpha_\theta(z) |z\rangle . \tag{C.1.13}$$

This equality explains why the output is a superposition. Unless $|\Phi_\theta\rangle$ equals one grid wave exactly, it has components along several vectors of the grid-wave basis. A unitary decoder preserves that linear expansion; it does not collapse or randomly choose one component.

C.1.iii Exact-grid and off-grid phases

If $\theta = k/M$ for an integer k, then $|\Phi_\theta\rangle = |w_k\rangle$, so the decoder produces $|k\rangle$ and measurement returns k with probability 1. The represented phase is exact: $k/M = \theta$.

If θ is not on the grid, the decoder instead produces

$$\sum_{z=0}^{M-1} \alpha_\theta(z) |z\rangle , \tag{C.1.14}$$

and measurement returns one value $\widetilde{z}$ according to

$$\Pr[\widetilde{z} = z \mid \theta] = |\alpha_\theta(z)|^2 . \tag{C.1.15}$$

Thus the phrase "PEA returns z" abbreviates two distinct steps:

$$\underbrace{\sum_z \alpha_\theta(z) |z\rangle}_{\text{unitary circuit output}} \xrightarrow{\text{measurement}} \underbrace{|\widetilde{z}\rangle}_{\text{one observed label}} . \tag{C.1.16}$$

The measured integer is not itself the phase; the phase estimate is $\widetilde{z}/M$.

C.1.iv Why the amplitudes peak near the true phase

For a candidate z, define its phase mismatch

$$\delta_z := \theta - \frac{z}{M}. \tag{C.1.17}$$

Then

$$\alpha_\theta(z) = \frac{1}{M} \sum_{j=0}^{M-1} e^{2\pi i j \delta_z}. \tag{C.1.18}$$

This is a sum of M unit arrows in the complex plane. Moving from branch j to $j+1$ rotates the next arrow by the fixed angle $2\pi\delta_z$. When z/M is close to θ modulo one turn, the mismatch modulo one is small, the arrows remain aligned, and their sum is large. For distant candidates they rotate through many directions and largely cancel. The controlled powers make the comparison sensitive because branch j accumulates the mismatch $j\delta_z$.

Let ℓ^{near} denote a nearest integer to $M\theta$, and let $z^{\text{near}} := \ell^{\text{near}} \bmod M$ be its measured label in $\{0, \ldots, M-1\}$. The amplitude formula is periodic in the integer label: $\alpha_\theta(z+M) = \alpha_\theta(z)$, since each summand acquires the factor $e^{-2\pi i j} = 1$. Thus the integer representative M describes the same measurement outcome as label 0. The unwrapped representative satisfies

$$|M\theta - \ell^{\text{near}}| \le \frac{1}{2} \quad \Longleftrightarrow \quad \left|\theta - \frac{\ell^{\text{near}}}{M}\right| \le \frac{1}{2M}. \tag{C.1.19}$$

This is why the peak is centered at a nearest finite-grid point, with labels wrapping from M to 0 when the phase is close to one full turn.

C.1.v The exact probability formula and the $4/\pi^2$ bound

The amplitude is a geometric sum. For $\delta_z \notin \mathbb{Z}$,

$$\alpha_\theta(z) = \frac{1}{M} \frac{1 - e^{2\pi i M \delta_z}}{1 - e^{2\pi i \delta_z}}, \qquad \Pr[\widetilde{z} = z \mid \theta] = \frac{1}{M^2} \frac{\sin^2(\pi M \delta_z)}{\sin^2(\pi \delta_z)}. \tag{C.1.20}$$

By periodicity, the probability of the measured label z^{near} can be calculated using its unwrapped representative ℓ^{near}. Set

$$\varepsilon := M\theta - \ell^{\text{near}}, \qquad |\varepsilon| \le \frac{1}{2}, \qquad \delta_{\ell^{\text{near}}} = \frac{\varepsilon}{M}. \tag{C.1.21}$$

For $\varepsilon \neq 0$, substitution gives

$$\Pr[\widetilde{z} = z^{\text{near}} \mid \theta] = \frac{1}{M^2} \frac{\sin^2(\pi\varepsilon)}{\sin^2(\pi\varepsilon/M)}. \tag{C.1.22}$$

The required lower bound now follows from two elementary sine inequalities. Because $0 \le \pi\,|\varepsilon| \le \pi/2$, concavity of sine on this interval gives

$$\sin(\pi\,|\varepsilon|) \ge 2\,|\varepsilon|\,. \tag{C.1.23}$$

For every $u \ge 0$, $\sin u \le u$, so

$$\sin(\pi\,|\varepsilon|\,/M) \le \frac{\pi\,|\varepsilon|}{M}. \tag{C.1.24}$$

Using these inequalities in the preceding probability yields the complete chain

$$\Pr[\widetilde{z} = z^{\text{near}} \mid \theta] \ge \frac{1}{M^2} \left(\frac{2\,|\varepsilon|}{\pi\,|\varepsilon|\,/M}\right)^2 = \frac{4}{\pi^2}. \tag{C.1.25}$$

If $\varepsilon = 0$, the phase lies exactly on the grid and the probability is 1, so the same lower bound holds in every case.

C.1.vi One complete state flow

For one fixed eigenstate, the complete process is

$$\begin{aligned} |0^m\rangle_P |\psi\rangle_T &\xrightarrow{H^{\otimes m}} \frac{1}{\sqrt{M}} \sum_j |j\rangle_P |\psi\rangle_T \\ &\xrightarrow{\text{controlled } U^j} \frac{1}{\sqrt{M}} \sum_j e^{2\pi i j\theta} |j\rangle_P |\psi\rangle_T \\ &\xrightarrow{\text{decoder}} \left(\sum_z \alpha_\theta(z) |z\rangle_P \right) |\psi\rangle_T \\ &\xrightarrow{\text{measure } P} |\widetilde{z}\rangle_P |\psi\rangle_T , \qquad \Pr[\widetilde{z} = z] = |\alpha_\theta(z)|^2 . \end{aligned} \tag{C.1.26}$$

The decoder creates the superposition over labels; measurement selects one label from it. The target remains the fixed input eigenstate throughout.

C.1.vii Connection to the discrete-logarithm construction

Question 4 takes $m = n$ and $N = p - 1$, so $M = 2^n > N$. For one fixed Fourier-wave state $|\chi_s\rangle$, the two phase estimations specialize to

$$U_g : \quad \theta_g = \frac{s}{N}, \qquad z_g^{\text{near}}(s) = \text{round}\left(\frac{Ms}{N}\right), \tag{C.1.27}$$

$$U_y : \quad \theta_y = \frac{sx \bmod N}{N}, \qquad z_y^{\text{near}}(s) = \text{round}\left(\frac{M(sx \bmod N)}{N}\right). \tag{C.1.28}$$

The additional issue in the full algorithm is that the easy target $|1\rangle$ is itself a superposition of many $|\chi_s\rangle$ states. That shared-target correlation is handled in the main report. The fixed-branch mechanism used there is now complete: controlled powers encode a phase wave, the decoder creates amplitudes over the finite labels, and measurement selects one observed label $\widetilde{z}$.

Main-text reference: Section 4.II.i.

C.2 Controlled Modular Multiplier Primer

The schoolbook reversible construction behind the Q4 cost bound. Main-text application: Section 4.III.i.

This primer expands the reversible-multiplier construction in Question 4. I specify the controlled-addition primitive, prove the accumulator invariant, and show how the auxiliary register and range flag return to zero. The final count records the gate cost, reusable workspace, and classical constants.

C.2.i Target operation and controlled powers

Let p be an n-bit prime and let $M = 2^n$. For a known $c \in \mathbb{Z}_p^*$, define the full-register unitary

$$U_c\,|a\rangle = \begin{cases} |ca \bmod p\rangle\,, & 1 \le a \le p-1, \\ |a\rangle\,, & a = 0 \text{ or } p \le a < M. \end{cases} \tag{C.2.1}$$

This permutes the valid nonzero residues and fixes every outside label. Phase estimation requires $U_c^{2^j}$ for $0 \le j < n$. Repeated multiplication acts as $|a\rangle \mapsto \left|c^{2^j} a \bmod p\right\rangle$ on valid labels, giving the full-register identity

$$c_j := c^{2^j} \bmod p, \qquad U_c^{2^j} = U_{c_j}. \tag{C.2.2}$$

Repeated squaring computes $c_0 = c$ and $c_{j+1} = c_j^2 \bmod p$ classically, so each stage uses one multiplier by c_j.

C.2.ii The controlled modular-addition primitive

Fix j and write the unknown quantum input in binary:

$$a = \sum_{k=0}^{n-1} a_k 2^k, \qquad a_k \in \{0,1\}. \tag{C.2.3}$$

For each input position k, precompute the hardwired constant

$$d_{j,k} := 2^k c_j \bmod p. \tag{C.2.4}$$

The reversible building block adds a known constant modulo p, controlled by one quantum bit:

$$\mathrm{CADD}_d^{(p)}\,|b\rangle\,|z\rangle = \begin{cases} |b\rangle\,|z + bd \bmod p\rangle\,, & 0 \le z < p, \\ |b\rangle\,|z\rangle\,, & p \le z < 2^n. \end{cases} \tag{C.2.5}$$

Here b is not measured. If $b = 0$, the circuit adds zero; if $b = 1$, it adds d. By linearity the same unitary works coherently when b is in superposition.

A ripple-carry addition, conditional subtraction of p, and uncomputation of the carries implement this primitive with $O(n)$ elementary gates and $O(n)$ reusable arithmetic ancillas [12].

C.2.iii Why the bitwise additions multiply by c_j

Start an n-qubit accumulator in $|0\rangle$. For $k = 0, 1, \ldots, n-1$, apply $\mathrm{CADD}_{d_{j,k}}^{(p)}$ controlled by the input bit a_k. Let z_r denote the accumulator after the first r input bits. The invariant is

$$z_r \equiv \sum_{k=0}^{r-1} a_k d_{j,k} \equiv c_j \sum_{k=0}^{r-1} a_k 2^k \pmod p. \tag{C.2.6}$$

It holds initially because the empty sum and z_0 are both zero. If it holds after r bits, the next controlled addition changes the accumulator by $a_r d_{j,r}$, so

$$z_{r+1} \equiv z_r + a_r d_{j,r} \equiv \sum_{k=0}^{r} a_k d_{j,k} \equiv c_j \sum_{k=0}^{r} a_k 2^k \pmod p. \tag{C.2.7}$$

Thus the invariant holds for every r. At $r = n$ it gives

$$z_n \equiv \sum_{k=0}^{n-1} a_k d_{j,k} \equiv c_j \sum_{k=0}^{n-1} a_k 2^k \equiv c_j a \pmod p. \tag{C.2.8}$$

The forward circuit has therefore produced the multiply-accumulate map

$$|a\rangle|0\rangle \longmapsto |a\rangle\left|c_j a \bmod p\right\rangle. \tag{C.2.9}$$

C.2.iv Cleaning the input register

The preceding map preserves the original input. To obtain a clean in-place multiplier, swap the two registers and run the inverse construction using $c_j^{-1} \bmod p$, which exists because $c_j \in \mathbb{Z}_p^*$. On valid inputs,

$$|a\rangle|0\rangle \longmapsto |a\rangle\left|c_j a\right\rangle \longmapsto \left|c_j a\right\rangle|a\rangle \longmapsto \left|c_j a\right\rangle\left|a - c_j^{-1}(c_j a)\right\rangle = \left|c_j a\right\rangle|0\rangle, \tag{C.2.10}$$

where every arithmetic expression is modulo p. The inverse cleanup uses the additional hardwired constants

$$e_{j,k} := 2^k c_j^{-1} \bmod p. \tag{C.2.11}$$

There are n controlled modular additions in the forward accumulation and n controlled modular subtractions in the inverse cleanup. The swap and the outer phase-estimation control change only constant factors.

C.2.v Fixing the outside basis labels

The arithmetic above is intended for $1 \le a \le p-1$. A reversible range flag is computed from the input and controls the forward accumulation, swap, and inverse cleanup. When the label is 0 or at least p, the flag is off and the whole circuit acts as the identity, exactly matching the full-space definition of U_{c_j}. The flag can be uncomputed from the output because multiplication by nonzero c_j preserves membership in $\{1, \ldots, p-1\}$, while every outside label was fixed.

C.2.vi Resource count

The $2n$ controlled additions and subtractions each cost $O(n)$ gates and reuse $O(n)$ arithmetic ancillas, giving

$$\text{one clean controlled } U_{c_j}: \qquad O(n^2) \text{ gates}, \qquad A(n) = O(n) \text{ ancillas}. \tag{C.2.12}$$

For each of the two bases $c = g, y$, there are n constants c_j and their inverses, together with n^2 forward constants $d_{j,k}$ and n^2 cleanup constants $e_{j,k}$. Repeated squaring, modular inversion, doubling, and writing the schoolbook gate descriptions require $O(n^3)$ classical bit operations in one-time compilation; later runs reuse the same circuits.

Main-text reference: Section 4.III.i.

APPENDIX D

Question 5: Quantum fingerprint preparation

D.1 Implementing the coherent lookup

This calculation supplies the gate-level construction behind the sender's lookup in Section 5.b.i. The sender knows z classically and computes the table $u \mapsto f_z(u)$ using the field operations of $\mathbb{F}_q$, where $q = 2^a$. Let t_u be the a-bit basis label of $f_z(u)$. These qa classical bits specify which gates appear in the circuit; they are not loaded into qa qubits. The required action is

$$L_z : \quad |u\rangle_{\text{arg}} |v\rangle_{\text{val}} \left|0^{O(a)}\right\rangle_{\text{work}} \longmapsto |u\rangle_{\text{arg}} |v \oplus t_u\rangle_{\text{val}} \left|0^{O(a)}\right\rangle_{\text{work}}, \tag{D.1.1}$$

where $\oplus$ is bitwise XOR on the stored labels.

To implement one row s, first apply X gates to the address positions where the label of s has a zero. The transformed address is all ones exactly when $u = s$. A chain of Toffoli gates computes its AND into a flag qubit, using $O(a)$ gates and clean scratch qubits. For every one-bit in t_s, apply a CNOT from this flag to the corresponding value qubit: at most a CNOTs XOR the row's value into val when the flag is one. Finally, reverse the Toffoli chain and the address X gates. This restores the address and every scratch qubit while retaining the change to val.

Thus one row uses $O(a)$ gates and $O(a)$ reusable workspace. On each basis address u, only its matching row can change the value register; processing all q rows realizes L_z with $O(qa)$ gates. By linearity, the same circuit preserves the superposition over addresses. This is the a-bit-output analogue of the coherent lookup used in Lecture 13.[10]

Main-text reference: Section 5.b.i, equation (5.b.3).

[10] Course Lecture 13, communication-complexity discussion, paragraph "Move 1 — Alice writes in her bit, by a coherent lookup."

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